ARHAM BIN TARIQ (M.Sc.)

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WORK EXPERIENCE

Research Assistant - ML Enhanced Sensors

Fraunhofer EMFT

July 2023 – August 2025 Munich

- Collected, cleaned, and annotated multimodal sensor time-series data, and conducted exploratory feature analysis.
- Designed, trained, and evaluated deep learning models using diverse performance metrics, leveraging GPU clusters for large-scale experimentation.
- Optimised models to reduce latency, memory footprint, and power consumption, while maintaining performance.
- Deployed optimised models on STM32 microcontrollers using TensorFlowLite and STM32 Cube AI, enabling real-time inference on embedded systems.

Research Assistant - Edge AI Systems

Fraunhofer EMFT

October 2021 – June 2023 Munich

- Collected, preprocessed, and labelled time-series sensor data from medical infusion pumps to detect air bubbles during liquid flow.
- Designed and optimised neural networks for time-series classification, including automated hyperparameter tuning.
- Applied pruning and quantisation to compress trained models, and deployed them on STM32 microcontrollers using C/C++, integrating with hardware components for real-time inference.
- Developed a systematic evaluation framework to assess accuracy, latency, and memory usage of deployed models; results contributed to a conference publication.

Research Assistant - Deep Learning for Computer Vision

Institute for Photonic Systems - Hochschule Weingarten

December 2020 – April 2021 Weingarten

- Built the deep learning module for an app to predict shoe size from foot image contours.

Intern - Software Architecture Development

ZF Friedrichshafen AG

August 2019 – February 2020 Friedrichshafen

- Developed ECU modules for automated software integration, flash process, and runtime analysis.

EDUCATION

M.Sc. Robotics, Cognition, Intelligence

Technical University of Munich

October 2021 – July 2025

- ***Thesis:*** Event Detection in High-frequency Multivariate Time Series for Tiny Machine Learning
- ***Focus Areas:*** Efficient Machine Learning, Robust Machine Learning, Foundation Models, Optimisation Algorithms

B.Eng. Electrical Engineering and Information Technology

Ravensburg-Weingarten University of Applied Sciences

March 2017 – July 2021

- ***Thesis:*** Navigation Control for an Autonomous Mobile Robot using Deep Reinforcement Learning
- ***Focus Areas:*** Reinforcement Learning, Robot Motion Planning, Computer Vision, Real-time Systems

PROJECTS

1. Event Detection in High-frequency Multivariate Time Series for Tiny Machine Learning

- Proposed a frequency-domain representation pipeline by converting time-series sensor data into spectrograms and reducing their dimensionality via frequency-bucket averaging, achieving over 80% compression without loss of event-specific detail.
- Designed and trained compact CNN architectures on minimal spectrograms, reaching 93% classification accuracy with a 250 kB model suitable for real-time inference on 25 kHz data streams.
- Fine-tuned pre-trained models from both time and frequency domains (InceptionTime, PANNs) to evaluate cross-domain transfer learning and validate the generalisation capability of the proposed approach.

2. Navigating the Neural Maze: A Strategic Guide for Evaluating and Improving the Robustness of Deep Neural Networks

- Investigated robustness enhancement in DNNs through replay-based retraining on corner-case samples and KL divergence regularisation to maintain previously learned representations across sequential tasks.
- Extracted and analysed latent-space representations to assess model generalisation, applying t-SNE for dimensionality reduction and HDBSCAN for density-based clustering.
- Evaluated clustering quality using multi-metric analysis (Silhouette Score, Davies–Bouldin Index, Cluster Purity) to identify the most effective retraining strategy.

3. Navigation Control for an Autonomous Mobile Robot using Deep Reinforcement Learning

- Simulated and trained a mobile robot in ROS and Gazebo using the Deep Deterministic Policy Gradient (DDPG) algorithm for autonomous navigation and obstacle avoidance.
- Investigated the impact of reward function design and environmental (state) conditions on reinforcement learning performance through comparative experiments.

PUBLICATIONS

Conference Paper

- K. Axelsson, A. B. Tariq, G. Zerbib, M. Richter, and C. Kutter, “Self-sensing of piezoelectric actuators: Integrated bubble detection for reliable drug dosing,” in *MikroSystemTechnik Kongress*, 2023.

SKILLS

Python | C/C++ | Java | HTML | CSS | JavaScript

PyTorch | TensorFlow | TensorFlow Lite | NumPy | SciPy | ML Flow | DVC | Label Studio | TensorBoard

Linux | Docker | LaTeX | Git | ROS | Gazebo | STM32 Programming

LANGUAGES

English	Native
German	C1 (Fluent)
Swedish	A2 (Elementary)
Japanese	A1 (Beginner)

VOLUNTEERING

Mentor for International Students
TUM Department of Informatics

Mentor for High School Students
Rock Your Life München